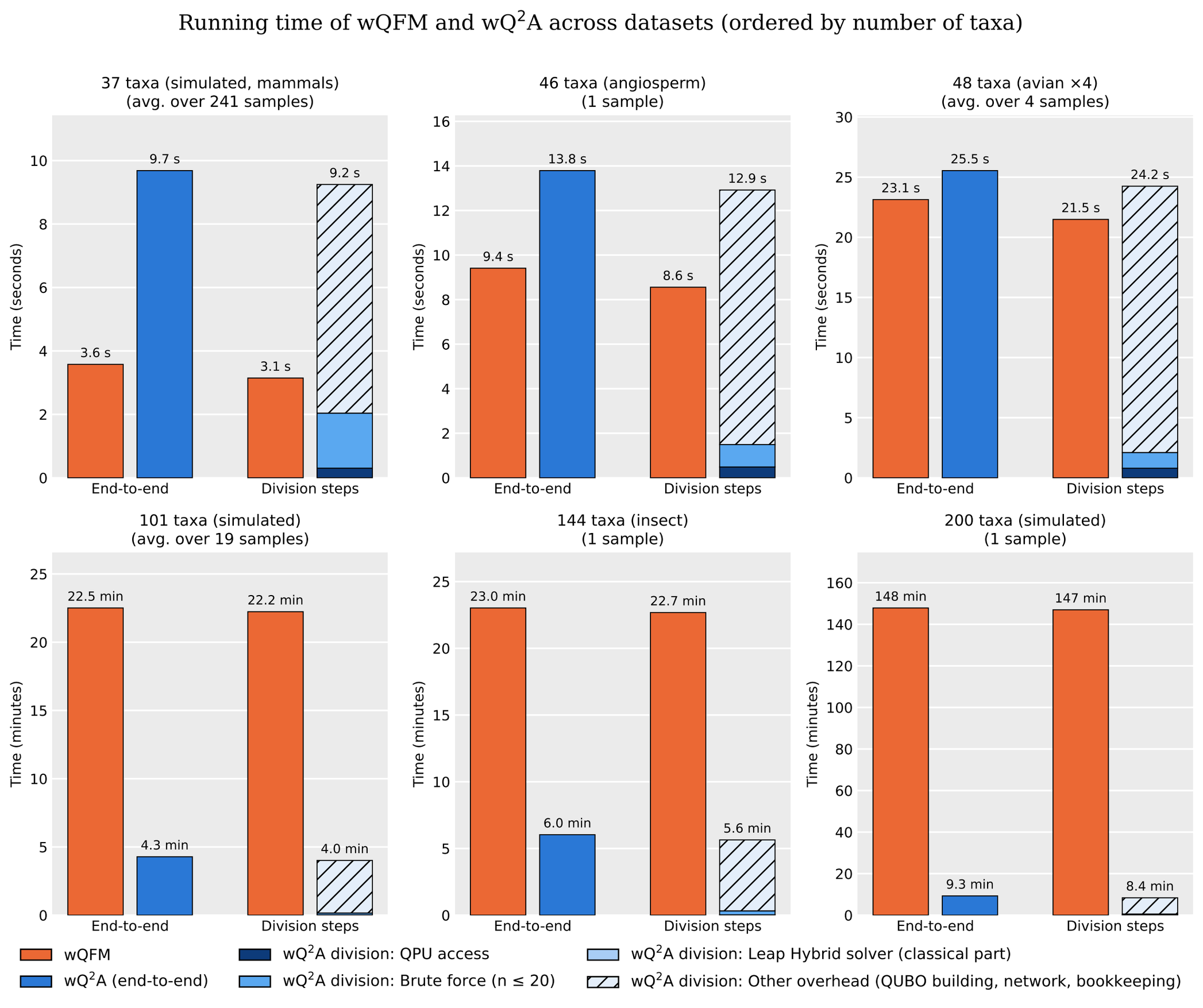
Running time of wQFM and wQ²A across datasets (ordered by number of taxa)
37 taxa (simulated, mammals)
(avg. over 241 samples)
46 taxa (angiosperm)
(1 sample)
48 taxa (avian ×4)
(avg. over 4 samples)
101 taxa (simulated)
(avg. over 19 samples)
144 taxa (insect)
(1 sample)
200 taxa (simulated)
(1 sample)
Time (seconds)
Time (minutes)
3.6 s
9.7 s
3.1 s
9.2 s
9.4 s
13.8 s
8.6 s
12.9 s
23.1 s
25.5 s
21.5 s
24.2 s
22.5 min
4.3 min
22.2 min
4.0 min
23.0 min
6.0 min
22.7 min
5.6 min
148 min
9.3 min
147 min
8.4 min
End-to-end
Division steps
wQFM
wQ²A (end-to-end)
wQ²A division: QPU access
wQ²A division: Brute force (n ≤ 20)
wQ²A division: Leap Hybrid solver (classical part)
wQ²A division: Other overhead (QUBO building, network, bookkeeping)